PAPER_781: M74 Phantom Galaxy — UQFF Grand Design Spiral Reference

Author: Daniel T. Murphy
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Abstract

M74 (NGC 628) is the "Phantom Galaxy," a nearly face-on grand-design spiral approximately 32 million light-years away ($z \approx 0.0022$) in Pisces. Known as one of the most symmetric spirals in the sky, M74 has been extensively studied by Hubble, Spitzer, and most recently JWST (which released stunning mid-infrared MIRI images in 2022 showing its spiral arms in extraordinary detail). With a moderate SFR ≈ 1–2 M⊙/yr and total mass $\sim 10^{11}$ M⊙, M74 is a textbook Sbc spiral. Under UQFF, standard disk rotation ($v = 10^5$ m/s) with typical galactic B-field yield g_M74 ≈ 1.053×10^{-3} m/s^2.

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1. Introduction

M74's nearly perfect face-on orientation (inclination ~5°) makes it an ideal case for tracing spiral structure and measuring the undisturbed rotation curve. JWST MIRI imaging in 2022 revealed star-forming clumps, dust filaments, and HII regions threading the symmetric spiral arms at previously unresolved spatial scales. The symmetric spiral structure indicates a stable, long-lived disk without major recent mergers. UQFF captures M74's regular disk rotation through standard $v = 10^5$ m/s coupling, with moderate M_sf = 0.045 reflecting its ~1.5 M⊙/yr SFR averaged over 5 Gyr of evolution.

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2. Master UQFF Gravity Equation

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g_M74(r, t) = (G × M) / r² × (1 + H(z)×t) × (1 + M_sf) × (1 + f_TRZ) + a_EM
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2.1 Parameters

Parameter	Symbol	Value	Source
Galaxy mass	M	10^{11} M⊙ = 1.989×10^{41} kg	Hubble/JWST
Disk radius	r	5×10^{20} m (~53 kly)	NED
SFR	—	1.5 M⊙/yr	JWST MIRI
Age	t	5×10^{9} yr = 1.578×10^{17} s	Hubble time
M_sf	—	0.045	UQFF SFR integral
Redshift	z	0.0022	Spectroscopic
v_EM	v	10^5 m/s	Disk rotation
B_EM	B	10^{-5} T	Galactic field
f_TRZ	—	0.04	UQFF

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3. Long-Form Derivation

Step 1: Base Gravitational Term

$g\_grav = 6.6743e-11 \times 1.989e41 / (5e20)^2 = 5.311e-11\ m/s^2$

Step 2: Cosmic Expansion Factor

$H(z) = 2.269e-18\ s^{-1}$; $H(z)\times t = 2.269e-18 \times 1.578e17 = 0.358$; factor = 1.358

Step 3: SFR Mass Fraction

$M\_sf = 0.045$; $1 + M\_sf = 1.045$

Step 4: Time-Reversal Correction

$f\_TRZ = 0.04$; $1 + f\_TRZ = 1.04$

Step 5: Gravitational Total

$g\_grav\_total = 5.311e-11 \times 1.358 \times 1.045 \times 1.04 = 7.856e-11\ m/s^2$

Step 6: Aether EM Correction

$a\_EM = (1.602e-19 \times 1e5 \times 1e-5 / 1.673e-27) \times 11 \times 1e-12 = 1.053e-3\ m/s^2$

Step 7: Final Solution

$g\_M74 = 7.856e-11 + 1.053e-3 \approx 1.053e-3\ m/s^2$

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4. Physical Interpretation

M74's near-perfect symmetry and moderate SFR place it exactly in the standard UQFF Sbc spiral category. JWST MIRI imagery showing crisp HII regions distributed along symmetric arms confirms that M74 is in a quiescent, undisturbed star-formation mode. The consistent g_M74 = 1.053×10^{-3} m/s^2 result joins NGC 2841 (PAPER_775), NGC 6217 (PAPER_777), and NGC 7049 (PAPER_779) in affirming UQFF universality across nearby spiral morphologies.

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5. Conclusions

UQFF applied to M74 Phantom Galaxy yields g $\approx 1.053\times10^{-3}$ m/s^2. As JWST's 2022 MIRI showcase target, M74 provides a clean observational anchor for standard UQFF Sbc galaxy calculations at z = 0.0022.

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